

Matthew Choi's Research Progress under Lloyd Lobo

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1 Introduction

Designers develop matching networks to minimize signal reflection and thereby maximize power transfer in a circuit. Matching networks are crucial for applications such as audio amplifiers and RF systems. Efficient power transfer is considered to be a key factor in high-power image applications, especially when it involves the intensity of light on biological matter. A matching network was proposed to maximize power transfer to form optical waveguides. We demonstrate the performance of the matching network transducer in multiple environments by plotting the transducer load impedance as a function of frequency.

2 Signal Reflection

Signal reflection is the phenomenon where a portion of the signal traveling through a transmission medium is partially reflected back toward the source rather than being fully transmitted to the load. Signal reflection is caused by impedance mismatches, leading to power loss and signal distortion.

2.1 Maximum Power Theorem

Impedance mismatches lead to suboptimal power transfer, as explained by the maximum power theorem. The maximum power theorem states that to obtain maximum power from a power source with internal resistance, the resistance of the load must be equal to that of the source. Assume that a circuit has an AC power source with phasor magnitude of voltage $|V_s|$, fixed source impedance of Z_S , and load impedance of Z_L . We can assume that a current of magnitude $|I|$ occurs by dividing the magnitude of the source voltage by the magnitude of the total impedance of the circuit. The magnitude $|I|$ is therefore: $\frac{|V_s|}{|Z_S|+|Z_L|}$.

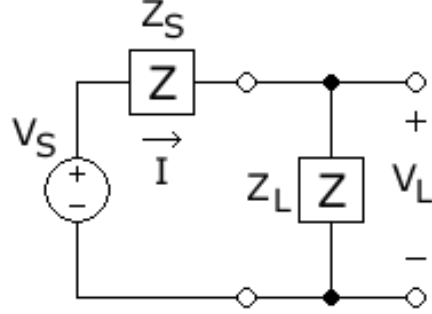


Figure 1: simplified reactive circuit

The average power P_L through the load is the square of current multiplied by the resistive(real) part R_L of the load impedance Z_L . Given R_S , R_L denote the resistances(real components), and X_S , X_L denote the reactances(imaginary components) of Z_S and Z_L respectively:

$$P_L = I_{RMS}^2 R_L = \frac{|I|^2 R_L}{2} = \frac{1}{2} \left(\frac{|V_S|}{|Z_S| + |Z_L|} \right)^2 = \frac{1}{2} \left(\frac{|V_S|^2 R_L}{(R_S + R_L)^2 + (X_S + X_L)^2} \right)$$

Given $|V_S|$, $|Z_S|$, $|Z_L|$, for power to reach the maximum, denominator: $(R_S + R_L)^2 + (X_S + X_L)^2$ must be minimized. Furthermore, since reactances X_S , X_L can be negative, they can cancel each other out, such that $X_S = -X_L$, resulting in denominator being revised as: $(R_S + R_L)^2$. Now, $P_L = \frac{1}{2} \frac{|V_S|^2 R_L}{(R_S + R_L)^2}$, and P_L is also $\frac{V_S^2}{R_S + R_L} R_L$, so $R_L = \frac{V_S^2}{R_S^2/R_L + 2R_S + R_L}$. Differentiating the denominator with respect to R_L showcases: $\frac{d}{dR_L} (R_S^2/R_L + 2R_S + R_L) = \frac{-R_S^2}{R_L^2} + 1$. If the derivative of the first derivative is zero, it indicates the minimum or maximum for P_L .

Therefore, $\frac{R_S^2}{R_L^2} = 1$, or $R_L = \pm R_S$.

Resistors are always positive, so $R_S = R_L$.

In total, for P_L to be maximized, $R_S = R_L$ and $X_S = -X_L$.

2.2 Isolators and Matching Networks

Two circuit designs that were proposed to minimize signal deflection and maximize power were isolated networks and matching networks.

2.2.1 Isolators and Circulators

Isolators allow signals to come only from one port and exit through another while blocking all signals that travel in the opposite direction. Circulators serve the same role as isolators, but have three ports such that port 1 leads to port 2, port 2 leads to port 3, and port 3 leads to port 1. Each port combination serves

as an individual isolator, as a result of eliminating signal reflection. However, neither design prioritizes maximum power transfer, absorbing only signals and not changing the load impedance.

2.2.2 Transformers

A transformer maximizes power transfer by matching the load impedance to the complex conjugate of the source impedance through magnetic coupling. The primary and secondary circuits are electrically isolated; energy is transferred via a time-varying magnetic field generated in the primary winding, which induces a corresponding current in the secondary winding. The impedance transformation ratio is determined solely by the turns ratio of the coils. However, transformer-based matching is not ideal for ultrasonic transducers operating at megahertz frequencies. This is due to practical limitations in core design and parasitic effects at high frequencies, as well as the transformer’s inability to accurately match both the resistive and reactive components of the transducer impedance.

2.2.3 Filters

Filters allow frequencies below or above a designated cutoff frequency range to pass while attenuating. Filter designs act as a matching network by filtering out unwanted frequency components to match the load and source. A filter with a cutoff near the operating frequency can match impedances while filtering out noise and addressing bandwidth concerns.

3 Designing the matching network

Our goal is for the ultrasonic transducer to oscillate efficiently across frequencies up to 3 MHz. This relatively low-frequency range is not optimal for transformer-based matching, as transformers do not effectively account for the reactive components of the load impedance. Furthermore, isolators and circulators do not improve power efficiency as they do not match the impedance between the source and load.

Filters can be categorized as either low-pass or high-pass filters, with low-pass filters filtering out higher frequencies, and vice versa for high-pass filters. In addition, filters can also be categorized by complexity between L, T and Pi networks. Filter effectiveness can be determined by the Q factor, a dimensionless parameter describing selectivity of resonance on a reactive system. The general equation of the Q factor is $Q = \sqrt{|\frac{R_L}{R_S} - 1|}$ for a simple network.

3.1 L network Filters

An L-network filter is a simple and widely used configuration for impedance matching and filtering, especially in RF and ultrasonic applications. It consists of just two reactive components—either an inductor and a capacitor—arranged

in an “L” shape. Depending on the placement of the components (series or shunt), an L-network can act as a low-pass or high-pass filter, while also transforming the source and load impedances to maximize power transfer. This dual functionality makes L-networks highly efficient for narrowband applications, such as driving ultrasonic transducers at a specific frequency. By selecting component values based on the desired frequency and impedance ratio, L-networks provide a compact and analytically tractable solution. However, their bandwidth is limited, the Q factor cannot be tuned, and they require the load and source impedances to be either both real or at least transformable into real values through conjugate matching techniques. Despite this, L-network filters remain a go-to solution due to their simplicity, low cost, and effectiveness at specific frequencies.

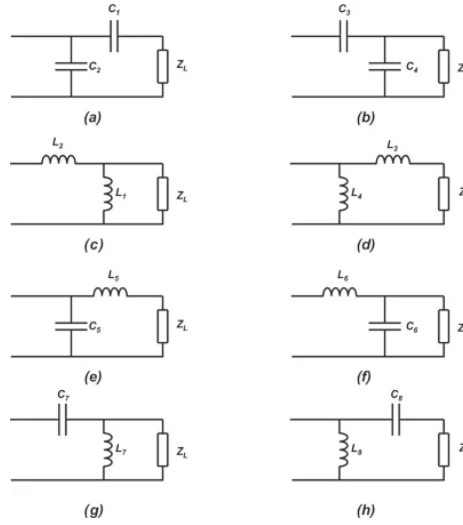


Figure 2: The 8 variations of the L network(a,b,c,d are band pass variants, while e,f are low pass and g h are high pass)

3.2 T network Filters

A T network filter is a complex variation of the L network filter that allows flexible control over the Q factor and bandwidth effects. The T network is composed of two reactive components in series, with a third shunt component in between. The T network can be viewed as two L networks placed in series, allowing the user to control both input and output impedances within the midpoint of the T network. This structure enables more complex impedance transformations and improved frequency selectivity compared to that of a single L-network. As a result, T networks are commonly used in applications requiring precise tuning and narrowband matching.

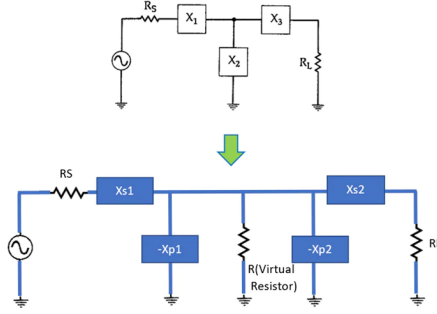


Figure 3: T network diagram where X_1, X_2, X_3 are reactive components

3.3 Pi network Filters

A Pi network is a three-element impedance matching circuit consisting of two reactive shunt components separated by a series reactive component, forming a shape resembling the Greek letter π . This configuration provides greater flexibility and improved control over impedance transformation compared to simpler L-networks as well as.

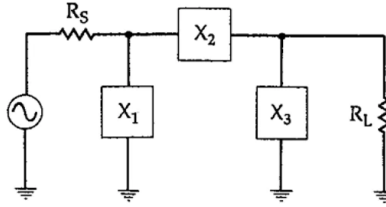


Figure 4: Pi network design configuration(X_1, X_2, X_3 are reactive components)

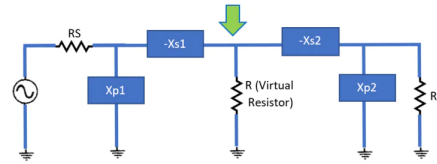


Figure 5: Pi network re-imagined with virtual resistor R

3.4 Designing network given absolute impedance and degree at designated cutoff

In order to calculate how reactive components convert the load impedance Z_L into conjugate source impedance Z_S^* , we need to identify Z_L , Z_S^* , and operating frequency.

3.4.1 Designing an L network

For an L network, since it only has one series and shunt component, there is no middle impedance point, indicating that the Q factor is already predetermined as $Q = \sqrt{|\frac{R_L}{R_S} - 1|}$. We then determine if we want the filter to be a low- or high-pass filter given the operating frequency and desired range. Since we are trying to convert the load impedance Z_L into the conjugate source impedance Z_S^* , we can trace the path using a Smith chart.

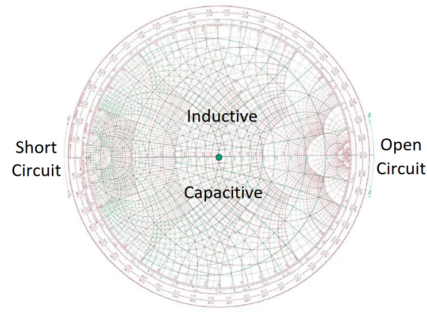


Figure 6: Sample Smith Chart

We can set Z_S as 50Ω , and naturally set Z_S^* as 50Ω due to the lack of reactive components. We can then draw out Z_L on to the chart by normalizing Z_L to Z_S , and then directly plotting on the chart. We then attempt to move towards the center of the chart by adding reactive components. The first reactive component will shift impedance to the real axis where reactance = 0 by traveling on the resistance circle. Afterwards, a second reactive component will shift the resistance of Z_L to Z_S^* .

Smith Chart tool link used for sanity checks

Reference link detailing on Smith Chart and how to calculate for L networks

Reference link to calculating for L networks

3.4.2 Designing an T/PI network

Designing either a T or PI network involves either finding the R that allows for a specific Q factor, or pre-determining the size of the virtual load. This can be done by converting the network into 2 separate L networks that share the same virtual load. We then calculate for the L networks that transform the impedance from Z_L to virtual load and from virtual load to Z_S^* .

Reference link detailing on constructing more complex filters

3.5 Finding Z_L with impedance analyzer

The tool we used to determine Z_L for the ultrasonic transducer was the 4294A Precision Impedance Analyzer, capable of accurately measuring the impedance from 40 to 110 MHz. In order to measure impedance, the following procedure is used in order.

1. Turn on Impedance Analyzer(labeled as LINE on the bottom left of the analyzer)
2. Attach either a test fixture or 2 sets of red and black jumper cables, each set is connected to the current and potential, and there is a HIGH and LOW set. For measuring the transducer by itself order does not matter, but for the matching network, make sure to have the HIGH side act as the voltage source and the LOW side be parallel to the load impedance. In addition, ensure ground cables are connected to each other.
3. Set oscillation to 500mV
4. Insert floppy disk into floppy disk reader built in the impedance analyzer
5. Ensure that measurement settings are set to $|Z| - \theta$ for polar impedance by pressing "Measure" and pressing the button with $|Z| - \theta$ directly to the left of it.
6. Set start frequency by pressing "start", and filling in respective frequency with number-pad on "entry" and metric prefixes.
7. Set stop frequency by pressing "stop", and filling in respective frequency with number-pad on "entry" and metric prefixes.
8. Calibrate Settings by pressing "Cal"
9. Check Adapter settings by pressing button with "Adapter", and then press button with the specific wire adapter being used(if uncertain like myself, set to NONE)
10. Begin Calibration by pressing return(back to the Cal page), and press button next to "Fixture Compn"
11. Run Open Calibration by having the two red jumper cables not touch to simulate open circuit behavior

12. Run Short Calibration by having the two red jumper cables clipped to simulate short circuit behavior
13. Run Load calibration by either connecting transducer to cables directly, or connecting the matching network with load impedance
14. Wait up to 2 minutes to let impedance analyzer calibrate(if settings do not seem accurate enough, go to Bw/Avg -i, Bandwidth and raise bandwidth to raise accuracy
15. Save file as txt by going to press button "Save" and set as "ASCII" file, and then rename the file to your comfort.
16. For more measurements, DO NOT undo calibration settings, and replace wiring orientation.
17. Remove floppy disk from Impedance Analyzer.

Impedance Analyzer Specifications

4 Findings

In April, we measured ultrasound measurements at varying frequencies from 50 to 300 kHz, as well as a singular sweep from 500 kHz to 15 MHz. The results indicated that the transducer load was mainly 50, with two exceptions at around 60 kHz and 2.78 MHz. Given that we wanted the transducer frequency to be at least 1 MHz, we swept the impedance using the same transducer from 2.5 to 3 MHz to see if peaks in absolute impedance were detected.

Given the nature of ultrasonic transducers, they exhibit both resonant and anti-resonant behavior within a small spectrum. Therefore, it is imperative to keep an eye out for points where the absolute magnitude of impedance is at minimum and peak.

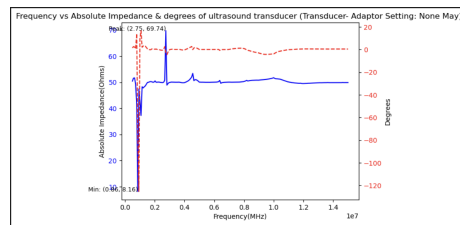


Figure 7: Anti resonant behavior shows up at around 2.74 MHz as shown by the peak of 73.4Ω

After finding a resonance frequency of 2.74 MHz and a maximum absolute load impedance of 73.4Ω , a low pass L network was initially designed(because I wanted to preserve as much bandwidth as possible, whether the filter is high or

low is unimportant due to broad frequency range). Since $Z_L > Z_S$, a series L and shunt C can be used as a low pass L network.

4.1 Precise Calculation process for L network(without smith chart)

1. Normalize the load impedance:

$$z_L = \frac{Z_L}{Z_0} = r_L + jx_L$$

2. Determine the location of the load on the Smith Chart:

- If $r_L > 1$, the load lies *inside* the $1 + jx$ circle.
- If $r_L < 1$, the load lies *outside* the $1 + jx$ circle.

3. Choose the network topology:

- For $r_L > 1$: Use a *series reactance* followed by a *shunt susceptance*.
- For $r_L < 1$: Use a *shunt susceptance* followed by a *series reactance*.

4. Compute matching components analytically:

- For $r_L > 1$, solve:

$$X = X_L \pm \sqrt{Z_S(R_L - Z_S)}$$

$$B = \pm \frac{\sqrt{Z_S(R_L - Z_S)}}{Z_S R_L}$$

- For $r_L < 1$, compute admittance:

$$Y_L = \frac{1}{Z_L}$$

and solve:

$$X = \pm \sqrt{Z_S \left(\frac{Z_S}{R_L} - 1 \right)}$$

$$B = \pm \frac{\sqrt{Z_S \left(\frac{Z_S}{R_L} - 1 \right)}}{Z_S R_L}$$

5. Convert X and B to lumped element values:

- For series reactance X :

$$L = \frac{X}{2\pi f} \quad (\text{for inductors}), \quad C = \frac{1}{2\pi f |X|} \quad (\text{for capacitors})$$

- For shunt susceptance B :

$$C = \frac{B}{2\pi f} \quad (\text{if capacitive}), \quad L = \frac{1}{2\pi f|B|} \quad (\text{if inductive})$$

6. **Verify match:** Ensure the input impedance is:

$$Z_{\text{in}} = Z_0 \quad \text{or} \quad \Gamma = 0$$

Using procedure above, the circuit that I developed had series inductor of $2\mu H$ and shunt capacitor of $520pF$, but when drawn in a Smith Chart, the Z_L shown on the chart had the same resistance, but inverted reactance, so capacitor had to be increased to $520pF$.

When testing the L network onto the transducer, we got an unexpected graph result, only to release that we were using the wrong transducer, as shown by the appearance of a resonance point very close to anti resonance at 2.78 MHz.

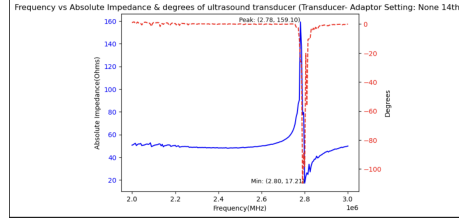


Figure 8: Frequency vs Impedance for transducer without L network

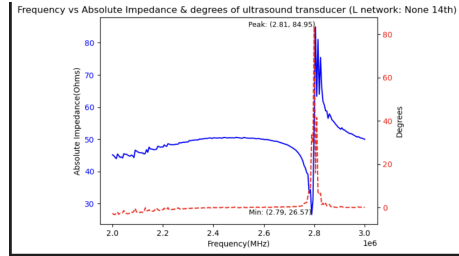


Figure 9: Frequency vs Impedance for transducer with L network

However, at the time, we were not sure if the issue was with the transducer conditions being optimal, such as wire tangling, and took more measurements to fully determine the actual impedance of the transducer before redesigning the matching network. I employed different transducer adaptor settings to check for drastic changes, to check for change before deciding to use "NONE" as the default setting.

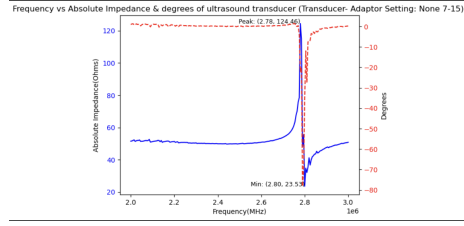


Figure 10: Frequency vs Impedance for transducer with NONE: transducer adaptor settings

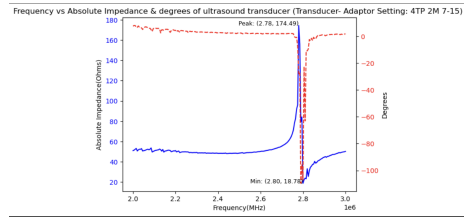


Figure 11: Frequency vs Impedance for transducer 4TP 2M transducer settings

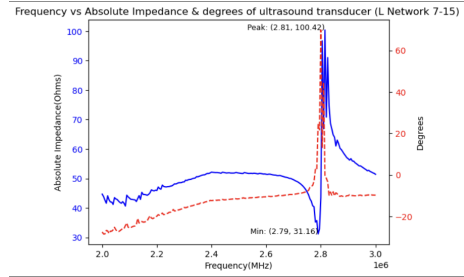


Figure 12: Frequency vs Impedance for transducer with previous L network

We redesigned the L network for absolute impedance of 124.46Ω and frequency of 2.78 MHz, resulting in a new design with series capacitor of 880 pF and 4.514 μ H. However, this still did not result optimal load impedance.

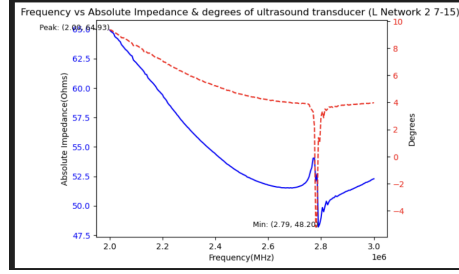


Figure 13: Frequency vs Impedance for transducer with new L network

We then tested more complex networks, and tried a Pi network with a quality factor of 3, leading to Z_L being essentially 50Ω the entire time in air. The network was composed of 1 shunt capacitor C_1 , followed by series inductor L_1 and second shunt capacitor C_2 . 1st shunt capacitor was 2 nF, the inductor was 3 uH, and the second capacitor was 1.5 nF.

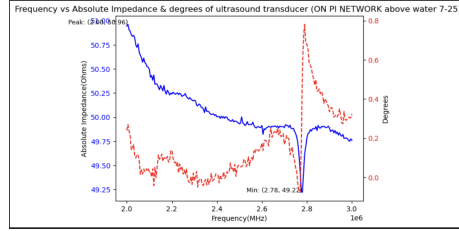


Figure 14: Frequency vs Impedance for transducer with PI network

Afterwards, we decided to take the Pi network and test the transducer in under-water conditions, and were expecting some change in the impedance. We used the same calibration settings for measurements of the transducer submerged and in normal setting, as well as with and without the PI matching network.

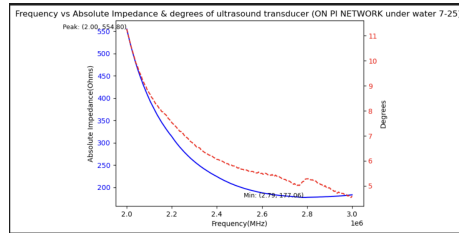


Figure 15: Frequency vs Impedance for submerged transducer with PI network

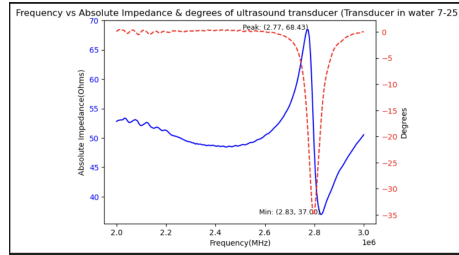


Figure 16: Frequency vs Impedance for submerged transducer

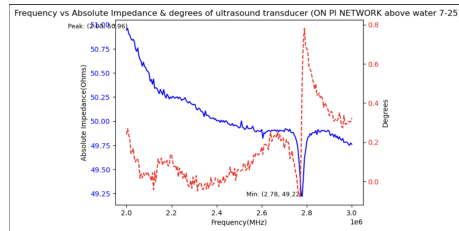


Figure 17: Frequency vs Impedance for transducer with PI network(not in water)

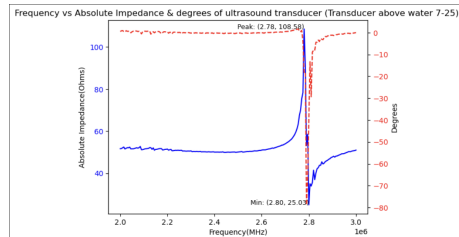


Figure 18: Frequency vs Impedance for transducer(not in water)

A few key comments in mind:

1. It may be far more beneficial to use the impedance analyzer to measure the reactance values before using the components on a breadboard
2. each reactive component is NOT perfect(the capacitors and inductors have a 10 percent margin of error)